Final 150 Descriptors Used for Classification

Table 1: Final features selected for model training

FeatureName	Description
SMR_VSA4	Van der Waals surface area related to molecular refractivity.
MaxPartialCharge	Maximum partial atomic charge in the molecule.
BCUT2D_MWHI	Topological descriptor based on eigenvalues of weighted adjacency matrix.
Kappa3	Third-order molecular shape index measuring complexity and branching.
charge.2.residue0	Partial or net atomic charge at specified residue or globally.
hydrophobicity_PRAM900101.3.residue0	Hydrophobicity score from AAIndex database for a residue/region.
SlogP_VSA4	Surface area of atoms in SlogP range indicating lipophilicity.
hydrophobicity_CASG920101.2.residue0	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_CASG920101.3.residue75	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_ARGP820101.Tr2332	Hydrophobicity score from AAIndex database for a residue/region.
solventaccess.1.residue0	Solvent accessibility of a residue in the peptide.
secondarystruct.1.residue0	Secondary structure propensity at the specified residue.
PEOE_VSA6	Electrostatic potential-derived van der Waals surface area.
hydrophobicity_ARGP820101.3.residue25	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_CASG920101.1.residue0	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_CASG920101.3.residue0	Hydrophobicity score from AAIndex database for a residue/region.
fr_NH1	Presence of primary amine functional group (-NH2).

solventaccess.3.residue0	Solvent accessibility of a residue in the peptide.
MinPartialCharge	Minimum partial atomic charge in the molecule.
NumAtomStereoCenters	Number of atomic stereocenters in the molecule.
polarizability.3.residue25	Polarizability value indicating electron cloud distortion potential.
charge.Tr1221	Partial or net atomic charge at specified residue or globally.
normwaalsvolume.3.residue0	Normalized van-der Waals volume of a residue.
hydrophobicity_ARGP820101.1.residue25	Hydrophobicity score from AAIndex database for a residue/region.
polarizability.3.residue0	Polarizability value indicating electron cloud distortion potential.
normwaalsvolume.3.residue25	Normalized van der Waals volume of a residue.
normwaalsvolume.3.residue75	Normalized van der Waals volume of a residue.
secondarystruct.2.residue50	Secondary structure propensity at the specified residue.
secondarystruct.3.residue100	Secondary structure propensity at the specified residue.
hydrophobicity_ARGP820101.3.residue0	Hydrophobicity score from AAIndex database for a residue/region.
MaxAbsPartialCharge	Maximum absolute value of partial atomic charge.
secondarystruct.1.residue50	Secondary structure propensity at the specified residue.
hydrophobicity_FASG890101.3.residue50	Hydrophobicity score from AAIndex database for a residue/region.
polarizability.3.residue75	Polarizability value indicating electron cloud distortion potential.
hydrophobicity_FASG890101.1.residue50	Hydrophobicity score from AAIndex database for a residue/region.
charge.1.residue75	Partial or net atomic charge at specified residue or globally.
EState_VSA4	Electrostatic state-related surface area feature.

secondarystruct.1.residue75	Secondary structure propensity at the specified residue.
hydrophobicity_PONP930101.3.residue75	Hydrophobicity score from AAIndex database for a residue/region.
polarizability.Tr1221	Polarizability value indicating electron cloud distortion potential.
normwaalsvolume.1.residue25	Normalized van der Waals volume of a residue.
polarity.1.residue75	Polarity score related to side chain dipole moments.
hydrophobicity_PRAM900101.3.residue25	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_ZIMJ680101.1.residue75	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_CASG920101.3.residue100	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_PRAM900101.1.residue0	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_FASG890101.1.residue0	Hydrophobicity score from AAIndex database for a residue/region.
charge.2.residue25	Partial or net atomic charge at specified residue or globally.
KK	Descriptor based on amino acid properties, RDKit, or AAIndex features.
hydrophobicity_FASG890101.1.residue75	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_ARGP820101.3.residue50	Hydrophobicity score from AAIndex database for a residue/region.
secondarystruct.2.residue100	Secondary structure propensity at the specified residue.
charge.1.residue100	Partial or net atomic charge at specified residue or globally.
secondarystruct.3.residue0	Secondary structure propensity at the specified residue.
hydrophobicity_ENGD860101.3.residue75	Hydrophobicity score from AAIndex database for a residue/region.
fr_C_O_noCOO	Carbonyl group count excluding carboxylic acid.
hydrophobicity_ZIMJ680101.3.residue0	Hydrophobicity score from AAIndex database for a residue/region.

solventaccess.3.residue75	Solvent accessibility of a residue in the peptide.
hydrophobicity_FASG890101.3.residue25	Hydrophobicity score from AAIndex database for a residue/region.
normwaalsvolume.1.residue75	Normalized van der Waals volume of a residue.
hydrophobicity_CASG920101.1.residue25	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_PRAM900101.2.residue0	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_PRAM900101.3.residue50	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_PRAM900101.1.residue75	Hydrophobicity score from AAIndex database for a residue/region.
VSA_EState8	Electrotopological state combined with surface area.
NumAmideBonds	Number of amide (peptide) bonds in the peptide.
hydrophobicity_CASG920101.2.residue50	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_PONP930101.1.residue50	Hydrophobicity score from AAIndex database for a residue/region.
charge.1.residue25	Partial or net atomic charge at specified residue or globally.
hydrophobicity_ENGD860101.1.residue50	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_ARGP820101.2.residue50	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_ZIMJ680101.1.residue50	Hydrophobicity score from AAIndex database for a residue/region.
fr_amide	Presence of amide functional group (e.g., -CONH2).
solventaccess.3.residue50	Solvent accessibility of a residue in the peptide.
hydrophobicity_PONP930101.1.residue75	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_CASG920101.2.residue25	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_FASG890101.1.residue100	Hydrophobicity score from AAIndex database for a residue/region.

secondarystruct.1.residue100	Secondary structure propensity at the specified residue.
hydrophobicity_PRAM900101.2.residue75	Hydrophobicity score from AAIndex database for a residue/region.
charge.1.residue0	Partial or net atomic charge at specified residue or globally.
hydrophobicity_CASG920101.1.residue50	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_PONP930101.2.residue75	Hydrophobicity score from AAIndex database for a residue/region.
normwaalsvolume.2.residue75	Normalized van der Waals volume of a residue.
normwaalsvolume.2.residue25	Normalized van der Waals volume of a residue.
Molar Extinction Coefficient (Oxidized Cysteines)	Absorbance coefficient for peptides with oxidized cysteine.
qed	Quantitative Estimate of Drug-likeness.
BCUT2D_MWLOW	Topological descriptor based on eigenvalues of weighted adjacency matrix.
hydrophobicity_ZIMJ680101.2.residue100	Hydrophobicity score from AAIndex database for a residue/region.
secondarystruct.3.residue50	Secondary structure propensity at the specified residue.
M	Descriptor based on amino acid properties, RDKit, or AAIndex features.
polarizability.1.residue50	Polarizability value indicating electron cloud distortion potential.
G	Descriptor based on amino acid properties, RDKit, or AAIndex features.
hydrophobicity_ZIMJ680101.2.residue75	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_ENGD860101.2.residue50	Hydrophobicity score from AAIndex database for a residue/region.
hydrophobicity_ARGP820101.3.residue100	Hydrophobicity score from AAIndex database for a residue/region.
PEOE_VSA4	Electrostatic potential-derived van der Waals surface area.
hydrophobicity_ARGP820101.2.residue100	Hydrophobicity score from AAIndex database for a residue/region.

polarizability.2.residue50	Polarizability value indicating electron cloud distortion potential.
hydrophobicity_PRAM900101.2.residue100	Hydrophobicity score from AAIndex database for a residue/region.
normwaalsvolume.1.residue50	Normalized van der Waals volume of a residue.
fr_NH2	Presence of secondary amine functional group.
hydrophobicity_ENGD860101.Tr2332	Hydrophobicity score from AAIndex database for a residue/region.
W	Descriptor based on amino acid properties, RDKit, or AAIndex features.
hydrophobicity_CASG920101.2.residue100	Hydrophobicity score from AAIndex database for a residue/region.
BCUT2D_CHGLO	Topological descriptor based on eigenvalues of weighted adjacency matrix.
KL	Descriptor based on amino acid properties
polarity.Tr2332	Polarity score related to side chain dipole moments.
WK	Descriptor based on amino acid properties
fr_benzene	Presence of benzene rings in molecular structure.
hydrophobicity_ARGP820101.Tr1331	Hydrophobicity score from AAIndex database for a residue/region.
charge.2.residue100	Partial or net atomic charge at specified residue or globally.
LF	Descriptor based on amino acid properties, RDKit, or AAIndex features.
BCUT2D_LOGPLOW	Topological descriptor based on eigenvalues of weighted adjacency matrix.
NumSaturatedHeterocycles	Count of saturated heterocyclic rings.
MaxEStateIndex	Descriptor based on amino acid properties
MaxAbsEStateIndex	Descriptor based on amino acid properties
KI	Descriptor based on amino acid properties

S.1	Descriptor based on amino acid properties, RDKit, or AAIndex features.
NumAromaticCarbocycles	Number of aromatic rings with only carbon atoms.
fr_Ar_N	Presence of aromatic nitrogen-containing ring.
AF	Descriptor based on amino acid properties, RDKit, or AAIndex features.
fr_phenol_noOrthoHbond	Phenol group presence (aromatic ring with -OH).
fr_guanido	Presence of guanidino functional group.
fr_COO	Carboxylic acid functional group (-COOH) presence.
ST	Describes the properties of serine and threonine, polar amino acids involved in phosphorylation.
LK	Describes leucine (hydrophobic) and lysine (basic, positively charged) amino acids.
WR	Describes tryptophan (aromatic, hydrophobic) and arginine (basic, positively charged) amino acids.
CT	Describes the composition, transition, and distribution patterns of amino acids in a peptide sequence.
KF	Represents the hydrophobicity of a peptide, influencing folding and solubility.
GL	Describes glycine (small, flexible) and leucine (large, hydrophobic) amino acids.
TY	Represents tyrosine, an aromatic amino acid involved in signaling and phosphorylation.
fr_COO2	Indicates the presence of a carboxyl group (-COOH), affecting charge and binding interactions.
GH	Describes glycine (flexible) and histidine (aromatic, proton donor/acceptor) amino acids.
GR	Describes glycine (flexible) and arginine (basic, positively charged) amino acids.
fr_Al_OH_noTert	Aliphatic alcohol group excluding tertiary OH.
SC	Descriptor based on amino acid properties, RDKit, or AAIndex features.
NumAromaticRings	Total number of aromatic rings in the peptide.

HN	Describes the properties of histidine) and asparagine (polar) amino acids.
QL	Describes glutamine and leucine (hydrophobic) amino acids.
ED	Represents glutamic acid (acidic) and aspartic acid (acidic) amino acids.
FG	Describes phenylalanine (aromatic, hydrophobic) and glycine (small, flexible) amino acids.
CN	Describes cysteine (sulfur-containing) and asparagine (polar, amide) amino acids.
fr_Al_COO	Indicates the presence of an aliphatic carboxyl group (-COOH), relevant to protein structure and interaction.
PR	Represents proline (helix-forming, cyclic) and serine (polar, hydroxyl group) amino acids.
FW	Describes the general hydrophobicity of amino acids, influencing peptide folding.
fr_phenol	Indicates the presence of a phenol group (-OH attached to an aromatic ring), relevant in aromatic stacking and interactions.
RP	Describes arginine (basic, positively charged) and proline (cyclic, helix-forming) amino acids.
TI	Represents threonine (polar, hydroxyl) and isoleucine (hydrophobic, branched-chain) amino acids.